Project RFM

Step 1 -- Download online retail file

https://archive.ics.uci.edu/dataset/352/online+retail

Step 2 – Convert into csv(easier to import)

Step 3 – Create the table with all columns as nvarchar(to avoid insert errors)

Step 4 – Use Bulk insert formula add proper location of the file

The file has been added

Data cleaning process begins

---Check the missing Customer Ids

```
use project_rfm
select
 count(*)
 from Online_Retail
 where CustomerID is null or CustomerID=' '
```

--Create a Clean Dataset

```
select *
into Online_retail_clean
from Online_Retail
where CustomerID is not null and CustomerID !=' '
```

--Check the Clean Dataset

```
select count(*)
from online_retail_clean
```

Removal of negative values

```
select *
into OnlineRetail_Final
from online_retail_clean
where quantity not like '-%'
```

Converting into required Data types

```
use project_rfm
SELECT
   InvoiceNo,
   StockCode,
   Description,
   TRY_CAST(Quantity AS INT) AS Quantity,
   TRY_CAST(InvoiceDate AS DATETIME) AS InvoiceDate,
   TRY_CAST(UnitPrice AS DECIMAL(10,2)) AS UnitPrice,
   TRY_CAST(CustomerID AS INT) AS CustomerID,
   Country
INTO Retail_Transformed_F
FROM OnlineRetail_Final
WHERE
   TRY_CAST(Quantity AS INT) IS NOT NULL
   AND TRY_CAST(UnitPrice AS DECIMAL(10,2)) IS NOT NULL
   AND TRY_CAST(CustomerID AS INT) IS NOT NULL;
```

Add a Total_Amount Column

i.e. Quantity * UnitPrice

```
alter table retail_transformed_f
add Total_amount decimal(12,2)
```

Then

```
update retail_transformed_f
set Total_amount = Quantity*Unitprice
```

Main Part i.e. The RFM

Recency

Frequency

Monetary

Code—

```
use project_rfm
select
        customerid,
        datediff(
                day,
                max(invoicedate),
                (select max(invoicedate) from retail_transformed_f)
                ) as Recency,
        Count(Distinct invoiceno) as Frequency,
        Sum(Total_Amount) as Monetary
        into Customer_RFM
from retail_transformed_f
group by customerid
```

SEGMENTATION

```
use project_rfm
select
        Customerid,
        Recency,
        Frequency,
        monetary,

        ntile(5) over(Order by recency asc) as R_Score,
        ntile(5) over (order by Frequency desc) as F_Score,
        ntile(5) over (order by Monetary desc) as M_Score
into RFM_Scores
from Customer_RFM
```

Create More Segments

```
use project_rfm
SELECT
  Customerid,
  Recency,
  Frequency,
  Monetary,
  R_Score,
  F_Score,
  M_Score,
  CASE
    when Recency is null then 'Others'

    when R_Score<=2 and F_Score>=4 and M_Score>=4 then 'Champions'
    when R_Score<=2 and F_Score>=3 then 'Loyal Customers'
```

```
        when R_Score<=3 and F_Score>=3 then 'Potential Loyalty'
        when R_Score>=4 and F_Score>=3 then 'At Risk'
        when R_Score>=4 and F_Score<=2 then 'Lost Customers'

        else 'Others'
    END as Customer_Segments
INTO Customer_segments
FROM RFM_Scores
```

Champions, Loyal, Potential Loyalty, At Risk, Lost Customers

Created according to their rfm scores

SUMMARIZATION CODE

```
select
            customer_segments,
            count(*) as Customers,
            sum(Monetary) as Revenue
from customer_segments
group by customer_segments
```